

CORE COMPUTER SCIENCE

Computer Networks

A visual reference on network types, OSI/TCP-IP, protocols & addressing

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- › LAN, VPN & IPv4 Addressing
- › OSI & TCP/IP Models
- › Application Layer Protocols
- › TCP vs. UDP & Handshakes
- › ARP, DHCP & ICMP
- › MAC Addressing & Subnetting
- › Firewalls & Transmission Types

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01 What Is a Network?

A **network** is an informally interconnected group or association of different entities — people, computers, radio stations, and so on. A **computer network** is a system of peripherals or computers interconnected with a standard communication channel to exchange information and data.

The **Internet** is a network of networks, connecting network-enabled devices worldwide so they can share data — which makes computer networking a core part of both everyday life and technical interviews.

1 How Are Network Types Classified?

Network types are classified by the geographic area they cover — from the range of a single device to the scale of the entire planet.

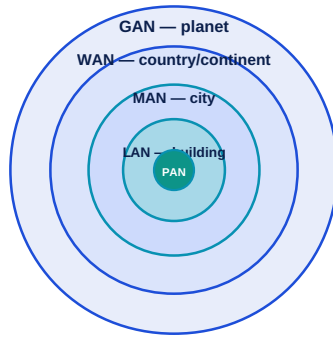


Fig 1 — Network types nest by geographic reach: $PAN \subset LAN \subset MAN \subset WAN \subset GAN$.

Distance	Region	Network Type
1 m	Square meter	Personal Area Network
10 m – 1 km	Room / Building / Campus	Local Area Network
10 km	City	Metropolitan Area Network
100 km – 1000 km	Country / Continent	Wide Area Network
10,000 km	Planet	The Internet (Global Area Network)

2 Types of Networks

Type	Description
PAN (Personal Area Network)	Lets devices connect over the range of a person. E.g. Bluetooth devices.
LAN (Local Area Network)	A privately owned network operating within/near a single building — home, office, or factory.
MAN (Metropolitan Area Network)	Connects and covers a whole city. E.g. TV cable connection across a city.
WAN (Wide Area Network)	Spans a large geographical area, often a country or continent. The Internet is the largest WAN.
GAN (Global Area Network)	Also known as the Internet — connects the globe using satellites; the "network of WANs".

02 LAN & VPN

3 LAN (Local Area Network)

LANs widely connect computers, laptops and consumer electronics, letting them share resources (printers, fax machines) and exchange information. LANs used by companies are called **enterprise networks**. There

are two types: **wireless LAN** (Wi-Fi, no wires) and **wired LAN** (LAN cable) — wireless is popular where running cable is difficult.



Fig 2 — Wireless LAN (Access Point + Wi-Fi) vs. wired LAN (Ethernet Switch + cable).

4 VPN (Virtual Private Network)

A VPN is a private WAN built on the internet. It creates a secured "tunnel" between networks over the public internet, letting a client connect to an organization's network remotely.

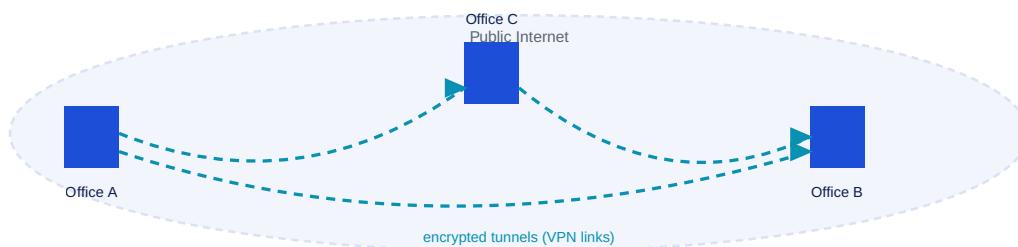


Fig 3 — A VPN links geographically separate offices through encrypted tunnels over the public internet.

5 Advantages of Using a VPN

- Connects offices in different geographical locations remotely — cheaper than dedicated WAN connections.
- Enables secure transactions and confidential data transfer between offices.
- Protects an organization's information against threats or intrusions using virtualization.
- Encrypts internet traffic and disguises the online identity.

6 Types of VPN

Access VPN

Connects remote mobile users and telecommuters — a low-cost alternative to dial-up or ISDN.

Site-to-Site VPN

A Router-to-Router VPN connecting branch office networks — common in large companies.

Intranet VPN

A Site-to-Site sub-type connecting remote offices under the same private-WAN accessibility policy.

Extranet VPN

A Site-to-Site sub-type connecting suppliers, customers and partners via dedicated connections.

03 Nodes, Links & Topology

7 Nodes and Links

Node

Any communicating device in a network — the point of intersection that can send/receive data. E.g. computers, printers, servers, modems.

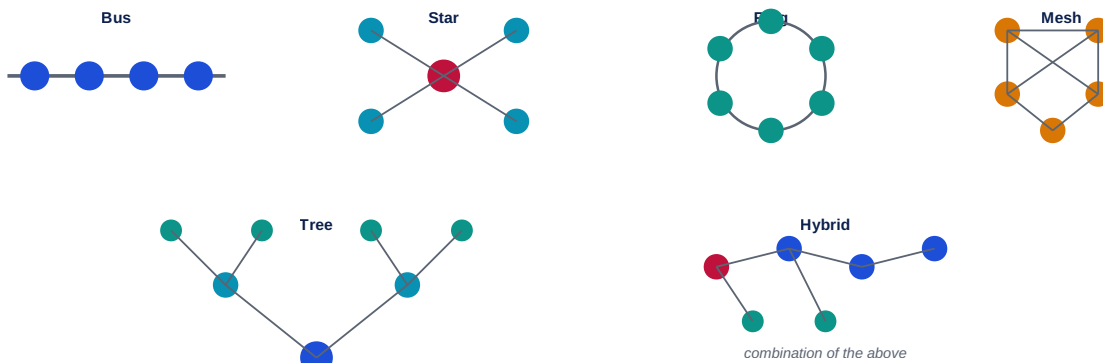
Link

The connectivity (edge) between two nodes — wired or wireless — including the protocol used to communicate.

8 Network Topology

Network topology is the physical layout of a network — how nodes connect via links, depicting connectivity between computers, devices and cables.

9 Types of Network Topology



combination of the above

Fig 4 — The six network topologies: Bus, Star, Ring, Mesh, Tree and Hybrid.

Topology	Key Traits
Bus	All nodes share one central cable (the bus). Good for small networks; if the cable fails, the whole network goes down.
Star	All nodes connect to one central node. Robust and easy to troubleshoot, but the whole network fails if the central node fails. Common in homes/offices.

Ring	Each node connects to exactly two neighbors, forming a ring. Rare — expensive and hard to install/manage; one damaged node breaks the network.
Mesh	Each node connects to one or many others. Robust — a failed link only disconnects that node — but rarely used due to install/management difficulty.
Tree	An "extended bus": smaller star networks connected to a single bus. If the main bus fails, the whole network is damaged.
Hybrid	Combines multiple topologies to cancel out individual weaknesses and combine their strengths.

04 IPv4 Addressing

10 IPv4 Address & Classes

An IP address is a 32-bit dynamic address for a node. An IPv4 address has 4 octets of 8 bits each, with each number up to 255. IPv4 classes are differentiated by the number of hosts supported, based on the first octet.

Class	Start Address	End Address	Usage
A	0.0.0.0	127.255.255.255	Large networks
B	128.0.0.0	191.255.255.255	Medium-size networks
C	192.0.0.0	223.255.255.255	Local Area Networks
D	224.0.0.0	239.255.255.255	Reserved for multicast
E	240.0.0.0	255.255.255.254	Study & R&D

11 Private & Special IP Addresses

Private Address: each class reserves specific IPs for private use only — non-routable, so they can't be used on the public internet.

Class	Private Start	Private End
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.31.255.255
C	192.168.0.0	192.168.255.255

SPECIAL ADDRESS

The range 127.0.0.1 – 127.255.255.255 is reserved for network testing — known as **loopback addresses**.

05 OSI Reference Model

12 The OSI Model

Open System Interconnection (OSI) is a network architecture model based on ISO standards, dealing with connecting systems open for communication with other systems. It has 7 layers, arrived at using these principles: create a new layer only when a different abstraction is needed; each layer has a well-defined function; each layer's function is chosen based on internationally standardized protocols.

13 The 7 OSI Layers

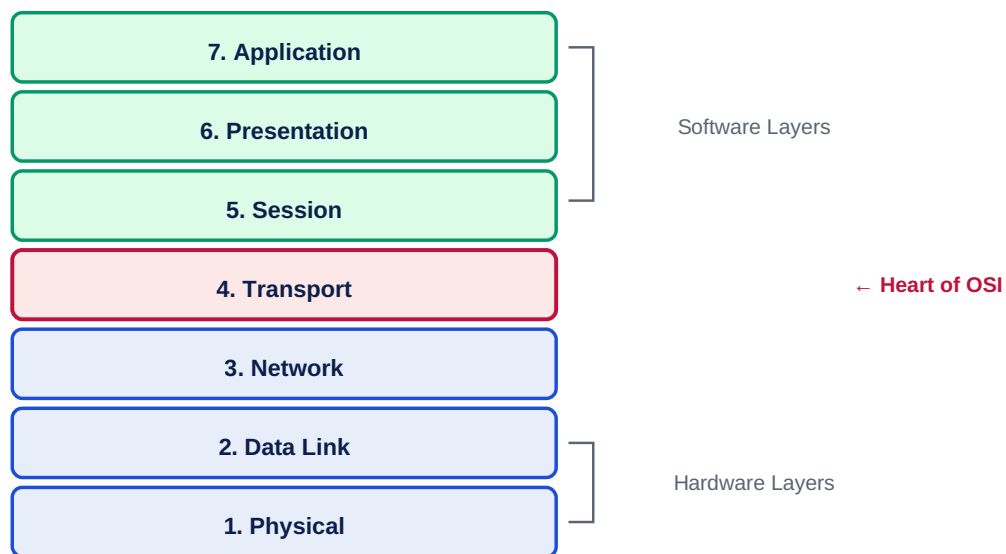


Fig 5 — The 7 OSI layers: software layers (5–7), hardware layers (1–2), and Transport as the "heart" bridging them.

Layer	Unit	Role
Physical	Bit	Transmits raw bits over a channel; picks the transmission mode (Simplex, Half-Duplex, Full-Duplex).
Data Link	Frame	Node-to-node delivery and error detection on the physical medium.
Network	Packet	Routing and logical addressing (IP) across networks.
Transport	Segment	End-to-end delivery, flow control and reliability — the "heart" of OSI.
Session	Data	Establishes, manages and terminates sessions between applications.
Presentation	Data	Translation, encryption and compression of data.
Application	Data	Provides network services directly to end-user applications.

06 TCP/IP Reference Model

14 The TCP/IP Model

A compressed version of OSI with only 4 layers, developed by the US Department of Defence (DoD). Named after its two core protocols: **TCP** (Transmission Control Protocol) and **IP** (Internet Protocol).

15 The 4 TCP/IP Layers

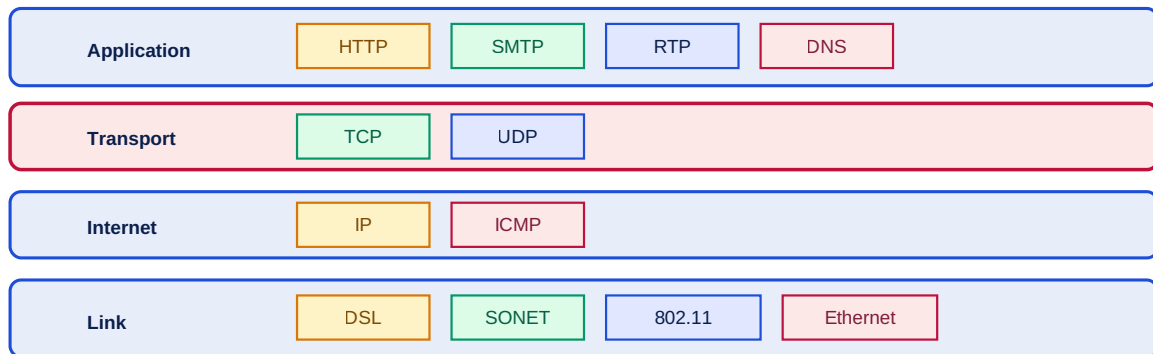


Fig 6 — TCP/IP's 4 layers, each mapped to representative protocols.

Layer	Role
Link	Chooses links (serial lines, classic Ethernet) to serve the connectionless internet layer.
Internet	The most important layer — delivers IP packets to their destination; holds the architecture together.
Transport	Similar to OSI's transport layer — lets peer entities carry on a conversation.
Application	Contains all higher-level protocols.

16 OSI vs. TCP/IP

OSI Reference Model	TCP/IP Reference Model
7-layer architecture	4-layer architecture
Fixed boundaries and functionality per layer	Flexible — no strict boundaries between layers
Low reliability	High reliability
Vertical layer approach	Horizontal layer approach

07 Application Layer Protocols

17 HTTP and HTTPS

HTTP (HyperText Transfer Protocol) defines the rules for transmitting information on the WWW. It's a *stateless protocol* — each command is independent of the previous one — built on TCP at the application layer, using port **80** by default.

HTTPS (HTTP Secure) is HTTP with SSL/TLS added on top, encrypting communication and securely identifying servers. It uses port **443** by default.

18 SMTP Protocol

SMTP (Simple Mail Transfer Protocol) sets the rules for server-to-server communication, letting software transmit emails over the internet. It supports End-to-End and Store-and-Forward methods, and always listens on port **25**.

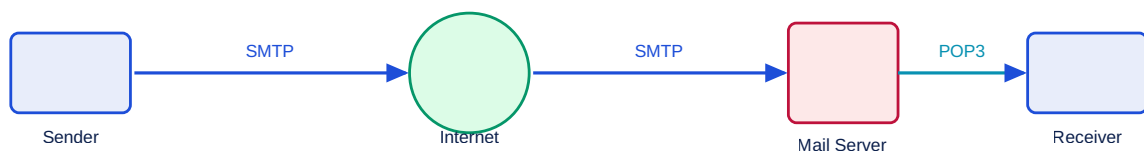


Fig 7 — SMTP carries mail from sender to mail server via the internet; POP3 delivers it to the receiver.

19 DNS

DNS (Domain Name System) is the Internet's device/service directory — a decentralized, hierarchical naming system that translates domain names into IPs (e.g. *example.com* → an IP address). It uses port **53** by default.

20 Router vs. Gateway

A **router** connects two or more network segments, directing traffic and transferring packets (web pages, emails, images) at the network layer. A **gateway** also routes traffic but can bridge two *dissimilar* networks — a router can only send data between similar networks.

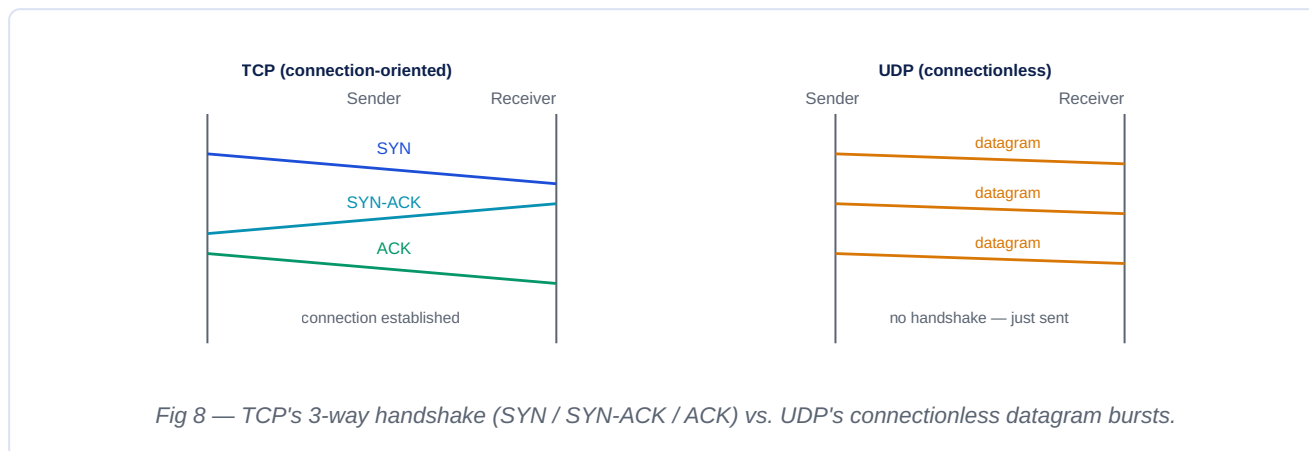
08 Transport Layer — TCP & UDP

21 TCP Protocol

TCP (Transmission Control Protocol) governs how a computer connects to the internet and transmits data. It creates a virtual, connection-oriented network and uses a **three-way handshake** to establish a connection, making it reliable.

22 UDP Protocol

UDP (User Datagram Protocol) is datagram-based, mainly used for multicasting and broadcasting. It skips the three-way handshake and error checking, using simple transmission — making it faster but less reliable.



23 TCP vs. UDP

TCP/IP	UDP
Connection-oriented	Connectionless
More reliable	Less reliable
Slower transmission	Faster transmission
Packet order preserved	Packet order not fixed, independent
Uses three-way handshake	No handshake
Heavier packets	Lightweight packets
Offers error checking	No error checking
HTTP, FTP, Telnet, SMTP, HTTPS	DNS, RIP, SNMP, RTP, BOOTP, TFTP

09 ICMP, DHCP & ARP

24 ICMP Protocol

ICMP (Internet Control Message Protocol) is a network layer protocol for error handling — used by routers to diagnose connection issues and confirm data reaches its destination. Uses port **7** by default.

25 DHCP Protocol

DHCP (Dynamic Host Configuration Protocol) is an application layer protocol that auto-configures devices on IP networks — auto-assigning IPs, subnet masks and DNS settings. Uses port **67** by default.

26 ARP Protocol

ARP (Address Resolution Protocol) is a network-level protocol that converts a logical address (IP address) into a device's physical address (MAC address) — used when devices communicate over the local network.

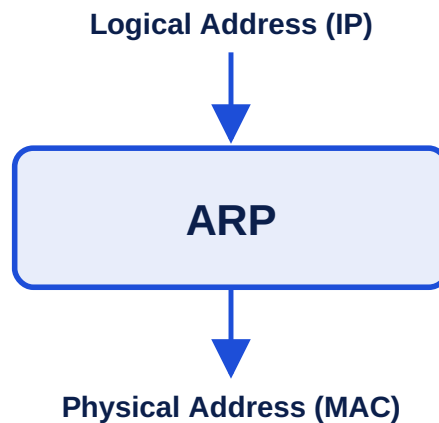


Fig 9 — ARP resolves a node's logical IP address into its physical MAC address.

27 FTP Protocol

FTP (File Transfer Protocol) is an application layer protocol for reliably and efficiently transferring files and data between hosts, including downloading from remote servers. Uses port **27** by default (commonly 20/21 in practice).

10 MAC Address & Subnetting

28 MAC Address & NIC

The **MAC address** (Media Access Control) is a 48-bit or 64-bit unique device identifier — also called the physical address — embedded in the **Network Interface Card (NIC)** used at the Data Link Layer. NIC is the hardware component through which a device connects to the network.

29 MAC vs. IP Address

MAC Address	IP Address
Media Access Control Address	Internet Protocol Address
6 or 8-byte hexadecimal number	4 bytes (IPv4) or 16 bytes (IPv6)
Embedded with NIC	Obtained from the network
Physical Address	Logical Address
Operates at Data Link Layer	Operates at Network Layer

30 Subnet

A **subnet** is a network inside a network, created through **subnetting** — dividing one network into smaller subnets. It improves routing efficiency, enhances security, and reduces the time to extract a host address from the routing table.

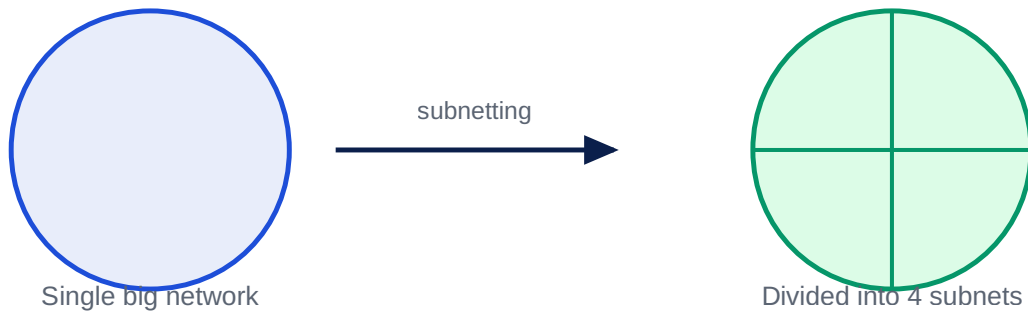


Fig 10 — Subnetting divides one large network into smaller, more manageable subnets.

11 Network Devices & Tools

31 Hub vs. Switch

Hub	Switch
Operates at Physical Layer	Operates at Data Link Layer
Half-Duplex transmission	Full-Duplex transmission
Connects Ethernet devices	Connects LAN devices
Less complex, cheaper	Intelligent and effective
No admin software support	Admin software support present
Speed up to 100 Mbps	Supports high speed (Gbps)
Less efficient — collisions likely	More efficient — collisions reduced

32 ipconfig vs. ifconfig

ipconfig	ifconfig
Internet Protocol Configuration	Interface Configuration

Both are used to view/configure network interfaces, get the TCP/IP summary, and change DHCP/DNS settings.

12 Firewalls & Transmission Types

33 Firewall

A **firewall** is a network security system that monitors incoming/outgoing traffic and blocks it per security policies — a wall between the public internet and private network devices. It can be hardware, software, or both.

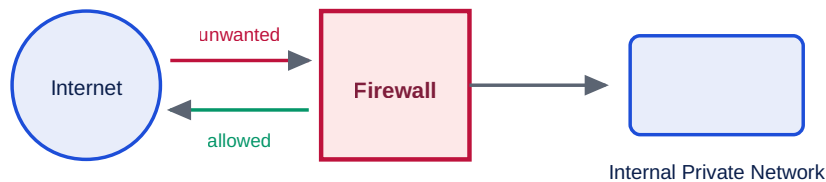


Fig 11 — A firewall filters traffic between the public internet and a private network.

34 Unicasting, Anycasting, Multicasting & Broadcasting

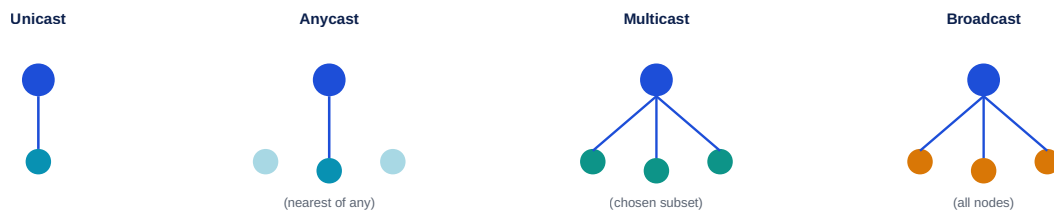


Fig 12 — Four transmission patterns: one-to-one, one-to-nearest, one-to-subset, one-to-all.

Unicasting

Message sent to a single node from the source. Commonly used to establish a new connection.

Anycasting

Message sent to any of the nodes from the source — often used to fetch content from the nearest server in a CDN.

Multicasting

Message sent to a chosen subset of nodes — used to send the same data to multiple receivers.

Broadcasting

Message sent to all nodes in the network. DHCP and ARP use broadcasting locally.

13 What Happens When You Enter google.com?

35 Step-by-Step Flow

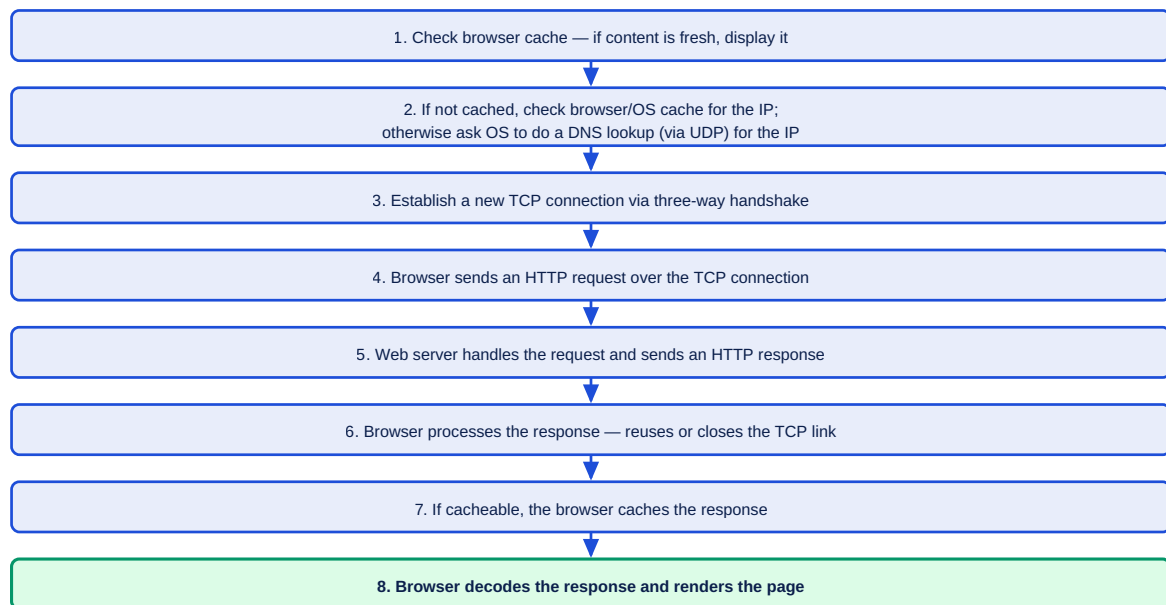


Fig 13 — The full round trip from typing a URL to a rendered web page.

FUN FACT

There is over 1.3 million kilometers of submarine optical fiber cable connecting the world to the Internet — enough to circle the earth more than 100 times.